



Top 62 CAT Profit and Loss Questions With Video Solutions

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Questions

Instructions

Directions for the following two questions: Shabnam is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%.

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, while a fall will entail a return of - 3%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of - 2.5%, while a fall will entail a return of + 2%.

Question 1

The maximum guaranteed return to Shabnam is

- A 0.25%
- B 0.10%
- C 0.20%
- D 0.15%
- E 0.30%

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Question (1)
The maximum guaranteed return to Shabnam is

A) 0.25%
B) 0.10%
C) 0.20%
D) 0.15%
E) 0.30%

0.10% option A
0.15% option B
0.30% option C
100-A

VIDEO SOLUTION

Question 2

What strategy will maximize the guaranteed return to Shabnam?

- A 100% in option A
- B 36% in option B and 64% in option C
- C 64% in option B and 36% in option C
- D 1/3 in each of the three options
- E 30% in option A, 32% in option B and 38% in option C

Directions for the following two questions: Shalini is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%.

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, while a fall will entail a return of -3%.

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Question (1)
The maximum guaranteed return to Shalini is

A) 0.25%
B) 0.50%
C) 0.20%
D) 0.15%
E) 0.30%

Handwritten notes: Option A, Option B, Option C, 100-A, 100-B, 100-C

VIDEO SOLUTION

Instructions

Directions for the following two questions:

Mr. David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs. 30. On the other hand, the cost, in rupees, of producing x units is $240 + bx + cx^2$, where b and c are some constants. Mr. David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by 66.67%. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximize the profit.

Question 3

How many units should Mr. David produce daily?

- A 130
- B 100
- C 70
- D 150
- E Cannot be determined

Directions for the following two questions:

Mr. David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs. 30. On the other hand, the cost, in rupees, of producing x units is $240 + bx + cx^2$, where b and c are some constants. Mr. David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by 66.67%. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximize the profit.

Handwritten notes: SP = 30N, CP = $240 + bN + cN^2$, $\frac{CP_{40}}{CP_{20}} = \frac{5}{3}$, $\frac{CP_{60}}{CP_{40}} = \frac{3}{2}$

VIDEO SOLUTION

Question 4

What is the maximum daily profit, in rupees, that Mr. David can realize from his business?

- A 620
- B 920
- C 840
- D 760
- E Cannot be determined

Directions for the following two questions:

Mr. David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs. 30. On the other hand, the cost, in rupees, of producing x units is $\frac{2x^3}{3} + bx + c$, where b and c are some constants. Mr. David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by 64.87%. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he produces. His objective is to maximize the profit.

Handwritten solution:

Num 0 $SP = 30N$

$CP = \frac{2N^3}{3} + bN + c$

$CP_{40} = \frac{5}{3}$ $CP_{60} = \frac{3}{2}$

$CP_{20} = 10$

$40b + c = 100$

$100c + 20b = 10$

[VIDEO SOLUTION](#)

Instructions

For the following questions answer them individually

Question 5

Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs. 3000, Rs. 2000 and Rs. 1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

Which of the following cannot be true?

- A At least two horses finished before Spotted
- B Red finished last
- C There were three horses between Black and Spotted
- D There were three horses between White and Red
- E Grey came in second

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Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs. 3000, Rs. 2000 and Rs. 1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

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B) Red finished last
C) There were three horses between Black and Spotted
D) There were three horses between White and Red
E) Grey came in second

1 →
2 →
3 → Red (3000)
4 →
5 →

1 → 4x
2 → 3x
3 → 2x
4 → 1x
5 → 0x

3000 → Red
2000 → White
1000 → Black
6000



VIDEO SOLUTION

Question 6

Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs. 3000, Rs. 2000 and Rs. 1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

Suppose, in addition, it is known that Grey came in fourth. Then which of the following cannot be true?

- A) Spotted came in first
- B) Red finished last
- C) White came in second
- D) Black came in second
- E) There was one horse between Black and White

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1 →
2 →
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4 →
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1 → 4x
2 → 3x
3 → 2x
4 → 1x
5 → 0x

3000 → Red
2000 → White
1000 → Black
6000



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Question 7

The owner of an art shop conducts his business in the following manner: every once in a while he raises his prices by $X\%$, then a while later he reduces all the new prices by $X\%$. After one such updown cycle, the price of a painting decreased by Rs. 441. After a second up-down cycle the painting was sold for Rs. 1,944.81. What was the original price of the painting?

- A Rs. 2,756.25
- B Rs. 2,256.25
- C Rs. 2,500
- D Rs. 2,000

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B) Rs. 2,256.25
C) Rs. 2,500
D) Rs. 2,000

$P - P(1+X)(1-X) = 441$
 $P - P(1-X^2) = 441$
 $PX^2 = 441$ (1)

$P(1+X)(1-X)(1+X)(1-X) = 1944.81$
 $P(1-X^2)^2 = 1944.81$

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Question 8

Gopal went to a fruit market with certain amount of money. With this money he can buy either 50 oranges or 40 mangoes. He retains 10% of the money for taxi fare. If he buys 20 mangoes, then the number of oranges he can buy is

- A 25
- B 18
- C 20
- D None of these

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A) 25
B) 18
C) 20
D) None of these

50 oranges = 40 mangoes = Rs. 200
 1 orange = Rs. 4
 20 mangoes = Rs. 200

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Question 9

A 20%

B 25%

C 40%

D 15%

 VIDEO SOLUTION

A Rs. 1.4 lakh

B Rs. 2 lakh

C Rs. 1 lakh

D Rs. 2.1 lakh

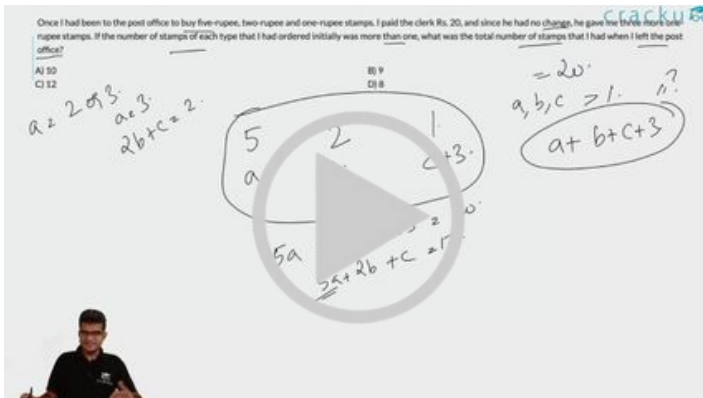
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VIDEO SOLUTION

Question 11

Once I had been to the post office to buy five-rupee, two-rupee and one-rupee stamps. I paid the clerk Rs. 20, and since he had no change, he gave me three more one-rupee stamps. If the number of stamps of each type that I had ordered initially was more than one, what was the total number of stamps that I had when I left the post office?

- A 10
- B 9
- C 12
- D 8



VIDEO SOLUTION

Question 12

The price of a Maruti car rises by 30% while the sales of the car come down by 20%. What is the percentage change in the total revenue?

- A -4%
- B -2%
- C +4%
- D +2%

VIEW SOLUTION



CAT Syllabus PDF



Question 13

I sold two watches for Rs. 300 each, one at the loss of 10% and the other at the profit of 10%. What is the percentage of loss(-) or profit(+) that resulted from the transaction?

- A (+)10

- B (-)1
- C (+)1
- D (-)10

I sold two watches for Rs. 300 each, one at the loss of 10% and the other at the profit of 10%. What is the percentage of loss(-) or profit(+) that resulted from the transaction?

A) (+)10
C) (+)1

B) (-)11
D) (-)10

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Total SP = 300 + 300 = 600

Total CP = $\frac{300}{0.9} + \frac{300}{1.1} = 300 + 300 \times \frac{10}{9} = 300 + 333.33 = 633.33$

Loss = 633.33 - 600 = 33.33

Percentage Loss = $\frac{33.33}{633.33} \times 100 = 5.26\%$

None of these

VIDEO SOLUTION

Question 14

After allowing a discount of 11.11%, a trader still makes a gain of 14.28%. At how many percentage above the cost price does he mark on his goods?

- A 28.56%
- B 35%
- C 22.22%
- D None of these

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A) 28.56%
C) 22.22%

B) 35%
D) None of these

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Let CP = Rs. 100

SP = 100 + 14.28% = 114.28

SP = MP - 11.11%

MP = $\frac{114.28}{0.8889} = 128.57$

Markup = 128.57 - 100 = 28.57%

28.56%

VIDEO SOLUTION

Question 15

A dealer buys dry fruits at Rs. 100, Rs. 80 and Rs. 60 per kilogram. He mixes them in the ratio 3 : 4 : 5 by weight, and sells at a profit of 50%. At what price per kilogram does he sell the dry fruits?

- A Rs. 80
- B Rs. 100

- C Rs. 95
- D None of these

A dealer buys dry fruits at Rs. 100, Rs. 80 and Rs. 60 per kilogram. He mixes them in the ratio 3 : 4 : 5 by weight, and sells at a profit of 50%. At what price per kilogram does he sell the dry fruits?

A) Rs. 80
B) Rs. 100
C) Rs. 95
D) None of these

VIDEO SOLUTION

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CAT Formula PDF

PDF

Question 16

A man earns $x\%$ on the first Rs. 2,000 and $y\%$ on the rest of his income. If he earns Rs. 700 from income of Rs. 4,000 and Rs. 900 from Rs. 5,000 of income, find $x\%$.

- A 20%
- B 15%
- C 25%
- D None of these

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A) 20%
B) 15%
C) 25%
D) None of these

VIDEO SOLUTION

Question 17

A yearly payment to the servant is Rs. 90 plus one turban. The servant leaves the job after 9 months and receives Rs. 65 and a turban. Then find the price of the turban.

- A Rs. 10
- B Rs. 15
- C Rs. 7.50
- D Cannot be determined

A yearly payment to the servant is Rs. 90 plus one turban. The servant leaves the job after 9 months and receives Rs. 65 and a turban. Then find the price of the turban.

A) Rs. 10
B) Rs. 15
C) Rs. 7.50
D) Cannot be determined

$90 + T \times \frac{9}{12} = 65 + T$

VIDEO SOLUTION

Question 18

Ravi invests 50% of his monthly savings in fixed deposits. Thirty percent of the rest of his savings is invested in stocks and the rest goes into Ravi's savings bank account. If the total amount deposited by him in the bank (for savings account and fixed deposits) is Rs 59500, then Ravi's total monthly savings (in Rs) is

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$FD = 50\%$ $Savings = 100\%$ 59500

VIDEO SOLUTION

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Question 19

If a seller gives a discount of 15% on retail price, she still makes a profit of 2%. Which of the following ensures that she makes a profit of 20%?

- A Give a discount of 5% on retail price.
- B Give a discount of 2% on retail price.

C Increase the retail price by 2%.

D Sell at retail price.

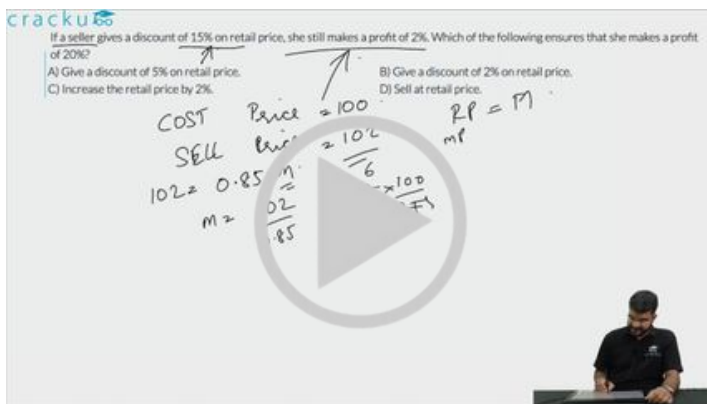
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If a seller gives a discount of 15% on retail price, she still makes a profit of 2%. Which of the following ensures that she makes a profit of 20%?

A) Give a discount of 5% on retail price.
B) Give a discount of 2% on retail price.
C) Increase the retail price by 2%.
D) Sell at retail price.

$\text{COST Price} = 100$
 $\text{Sell Price} = 100 \times 0.85 = 85$
 $102 = 0.85 \times \frac{100}{0.85}$
 $m = \frac{0.02}{0.85}$

$\text{RP} = \text{CP}$
 $m = \frac{0.02}{0.85}$



VIDEO SOLUTION

Question 20

Suppose, C1, C2, C3, C4, and C5 are five companies. The profits made by C1, C2, and C3 are in the ratio 9 : 10 : 8 while the profits made by C2, C4, and C5 are in the ratio 18 : 19 : 20. If C5 has made a profit of Rs 19 crore more than C1, then the total profit (in Rs) made by all five companies is

A 438 crore

B 435 crore

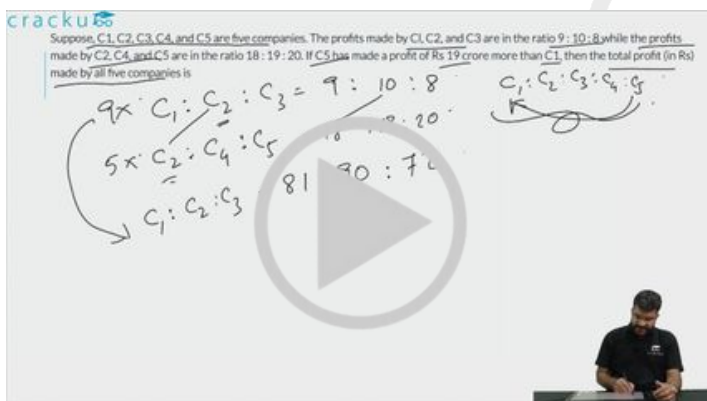
C 348 crore

D 345 crore

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$9 \times C_1 : C_2 : C_3 = 9 : 10 : 8$
 $5 \times C_2 : C_4 : C_5 = 18 : 19 : 20$
 $C_1 : C_2 : C_3 : C_4 : C_5$
 $81 : 90 : 72 : 190 : 200$



VIDEO SOLUTION

Question 21

In a market, the price of medium quality mangoes is half that of good mangoes. A shopkeeper buys 80 kg good mangoes and 40 kg medium quality mangoes from the market and then sells all these at a common price which is 10% less than the price at which he bought the good ones. His overall profit is

A 6%

B 8%

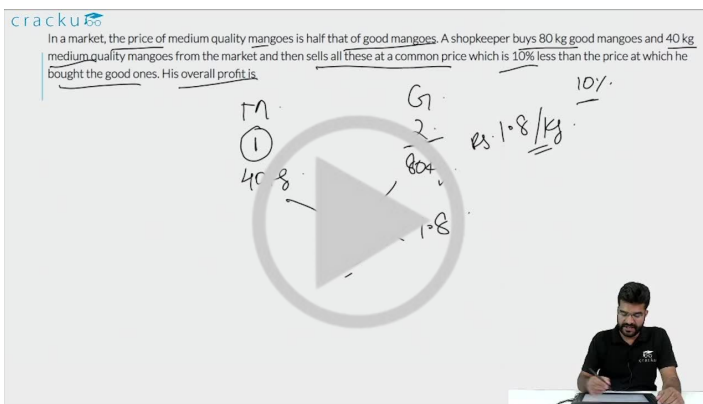
C 10%

D 12%

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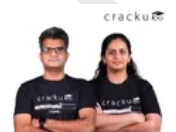
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Handwritten notes: 100 , 40×8 , 80×2 , $Rs. 1.8/kg$, 10% , 1.8



VIDEO SOLUTION

CAT Percentile Predictor



Question 22

If Fatima sells 60 identical toys at a 40% discount on the printed price, then she makes 20% profit. Ten of these toys are destroyed in fire. While selling the rest, how much discount should be given on the printed price so that she can make the same amount of profit?

A 30%

B 25%

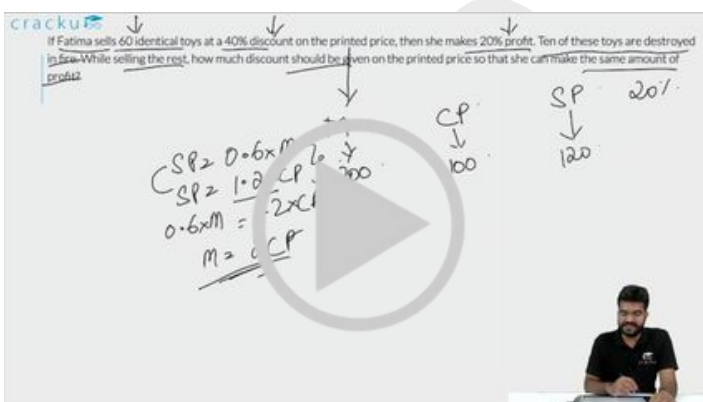
C 24%

D 28%

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If Fatima sells 60 identical toys at a 40% discount on the printed price, then she makes 20% profit. Ten of these toys are destroyed in fire. While selling the rest, how much discount should be given on the printed price so that she can make the same amount of profit?

Handwritten notes: $CP = 0.6 \times M$, $SP = 1.2 \times CP$, $0.6 \times M = 1.2 \times CP$, $M = 2 \times CP$, $CP = 100$, $SP = 120$, 20%



VIDEO SOLUTION

Question 23

The manufacturer of a table sells it to a wholesale dealer at a profit of 10%. The wholesale dealer sells the table to a retailer at a profit of 30%. Finally, the retailer sells it to a customer at a profit of 50%. If the customer pays Rs 4290 for the table, then its manufacturing cost (in Rs) is

- A 1500
- B 2000
- C 2500
- D 3000

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The manufacturer of a table sells it to a wholesale dealer at a profit of 10%. The wholesale dealer sells the table to a retailer at a profit of 30%. Finally, the retailer sells it to a customer at a profit of 50%. If the customer pays Rs 4290 for the table, then its manufacturing cost (in Rs) is

M → W → R → C

$1.1C$ $1.1C \times 1.3$ $1.1C \times 1.3 \times 1.5$

VIDEO SOLUTION

Question 24

Mayank buys some candies for Rs 15 a dozen and an equal number of different candies for Rs 12 a dozen. He sells all for Rs 16.50 a dozen and makes a profit of Rs 150. How many dozens of candies did he buy altogether?

- A 50
- B 30
- C 25
- D 45

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Mayank buys some candies for Rs 15 a dozen and an equal number of different candies for Rs 12 a dozen. He sells all for Rs 16.50 a dozen and makes a profit of Rs 150. How many dozens of candies did he buy altogether?

X Rs. 15/dozen
 X Rs. 12/dozen
 $2X$ Rs. 16.5/dozen

VIDEO SOLUTION

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CAT Score Calculator

Question 25

A shopkeeper sells two tables, each procured at cost price p , to Amal and Asim at a profit of 20% and at a loss of 20%, respectively. Amal sells his table to Bimal at a profit of 30%, while Asim sells his table to Barun at a loss of 30%. If the amounts paid by Bimal and Barun are x and y , respectively, then $(x - y) / p$ equals

- A 1
- B 1.2
- C 0.50
- D 0.7

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Shopkeeper

Amal

Bimal

Asim

Barun

1.2p

0.8p

1.56p

0.52p

VIDEO SOLUTION

Question 26

Mukesh purchased 10 bicycles in 2017, all at the same price. He sold six of these at a profit of 25% and the remaining four at a loss of 25%. If he made a total profit of Rs. 2000, then his purchase price of a bicycle, in Rupees, was

- A 6000
- B 8000
- C 4000
- D 2000

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A) 6000 B) 8000 C) 4000 D) 2000

2000

10C

6C

4C

1.25C

0.75C

2000

VIDEO SOLUTION

Question 27

The income of Amala is 20% more than that of Bimala and 20% less than that of Kamala. If Kamala's income goes down by 4% and Bimala's goes up by 10%, then the percentage by which Kamala's income would exceed Bimala's is nearest to

- A 31
- B 29
- C 28
- D 32

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The income of Amala is 20% more than that of Bimala and 20% less than that of Kamala. If Kamala's income goes down by 4% and Bimala's goes up by 10%, then the percentage by which Kamala's income would exceed Bimala's is nearest to

A 120
B 100
K $120/0.8 = 150$
 $150 \times 0.96 = 144$

VIDEO SOLUTION

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Question 28

Meena scores 40% in an examination and after review, even though her score is increased by 50%, she fails by 35 marks. If her post-review score is increased by 20%, she will have 7 marks more than the passing score. The percentage score needed for passing the examination is

- A 60
- B 80
- C 70
- D 75

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Meena scores 40% in an examination and after review, even though her score is increased by 50%, she fails by 35 marks. If her post-review score is increased by 20%, she will have 7 marks more than the passing score. The percentage score needed for passing the examination is

A) 60 B) 80
C) 70 D) 75

$a = 0.40a \times 1.5 = 0.6a$
 $0.6a + 35 = \text{passing}$
 $0.6a \times 1.2 = 0.72a$
 $0.72a = \text{passing} + 7$

VIDEO SOLUTION

Question 29

On selling a pen at 5% loss and a book at 15% gain, Karim gains Rs. 7. If he sells the pen at 5% gain and the book at 10% gain, he gains Rs. 13. What is the cost price of the book in Rupees?

- A 95
- B 85
- C 80
- D 100

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On selling a pen at 5% loss and a book at 15% gain, Karim gains Rs. 7. If he sells the pen at 5% gain and the book at 10% gain, he gains Rs. 13. What is the cost price of the book in Rupees?

$$\begin{array}{l} P \qquad B \\ -5\% \text{ of } P + 15\% \text{ of } B = 7 \\ 5\% \text{ of } P + 10\% \text{ of } B = 13 \end{array}$$


VIDEO SOLUTION

Question 30

A person spent Rs 50000 to purchase a desktop computer and a laptop computer. He sold the desktop at 20% profit and the laptop at 10% loss. If overall he made a 2% profit then the purchase price, in rupees, of the desktop is

A person spent Rs 50000 to purchase a desktop computer and a laptop computer. He sold the desktop at 20% profit and the laptop at 10% loss. If overall he made a 2% profit then the purchase price, in rupees, of the desktop is

$$\begin{array}{l} D + L = 50000 \\ 1.2D + 0.9L = 51000 \end{array}$$

$50000 \times \frac{102}{100}$



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Question 31

In a group of people, 28% of the members are young while the rest are old. If 65% of the members are literates, and 25% of the literates are young, then the percentage of old people among the illiterates is nearest to

- A 62
- B 55
- C 59
- D 66

In a group of people, 28% of the members are young while the rest are old. If 65% of the members are literates, and 25% of the literates are young, then the percentage of old people among the illiterates is nearest to

A) 62 B) 55
C) 59 D) 66

Total = 10000
Young = 2800
Old = 7200
Literates = 6500
Illiterates = 3500
Young = 6500 * 0.25 = 1625
Old = 3500 - 1625 = 1875

Percentage of old people among the illiterates = $\frac{1875}{3500} \times 100 = 53.57\%$

The nearest option is B) 55

VIDEO SOLUTION

Question 32

In the final examination, Bishnu scored 52% and Asha scored 64%. The marks obtained by Bishnu is 23 less, and that by Asha is 34 more than the marks obtained by Ramesh. The marks obtained by Geeta, who scored 84%, is

- A 357
- B 417
- C 439
- D 399

In the final examination, Bishnu scored 52% and Asha scored 64%. The marks obtained by Bishnu is 23 less, and that by Asha is 34 more than the marks obtained by Ramesh. The marks obtained by Geeta, who scored 84%, is

Total Marks = 100x
Bishnu = 52x = R - 23
Asha = 64x = R + 34

Solving for x:
52x = R - 23
64x = R + 34
Subtracting the first equation from the second:
12x = 57
x = 4.75

Geeta's marks = 84% of 100x = 84 * 4.75 = 399

VIDEO SOLUTION

Question 33

A man buys 35 kg of sugar and sets a marked price in order to make a 20% profit. He sells 5 kg at this price, and 15 kg at a 10% discount. Accidentally, 3 kg of sugar is wasted. He sells the remaining sugar by raising the marked price by p percent so as to make an overall profit of 15%. Then p is nearest to

- A 22
- B 35
- C 25
- D 31

cracku

A man buys 35 kg of sugar and sets a marked price in order to make a 20% profit. He sells 5 kg at this price, and 15 kg at a 10% discount. Accidentally, 3 kg of sugar is wasted. He sells the remaining sugar by raising the marked price by p percent so as to make an overall profit of 15%. Then p is nearest to

Total CP = 3500

CP of 1 kg of sugar = Rs. 100

SP = 120 × 0.9 = 108

MP = Rs. 120

5 × 120 = 600

15 × 108 = 1620

3 × 0 = 0

12 ×

VIDEO SOLUTION

cracku

CAT Daily Targets

Step by step, heading towards 99+ percentile

Question 34

Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been

- A 50
- B 60
- C 54
- D 55

cracku

Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been

12 (12)

8 ← 0.8M

4 ← 0.6M

0.8M

0.6M

6.4M + 2.4M = 8.8M

M = 8.8

8.8

8.8

0.8M × 0.75 = 0.6M

VIDEO SOLUTION

Question 35

Amal purchases some pens at ₹ 8 each. To sell these, he hires an employee at a fixed wage. He sells 100 of these pens at ₹ 12 each. If the remaining pens are sold at ₹ 11 each, then he makes a net profit of ₹ 300, while he makes a net loss of ₹ 300 if the remaining pens are sold at ₹ 9 each. The wage of the employee, in INR, is

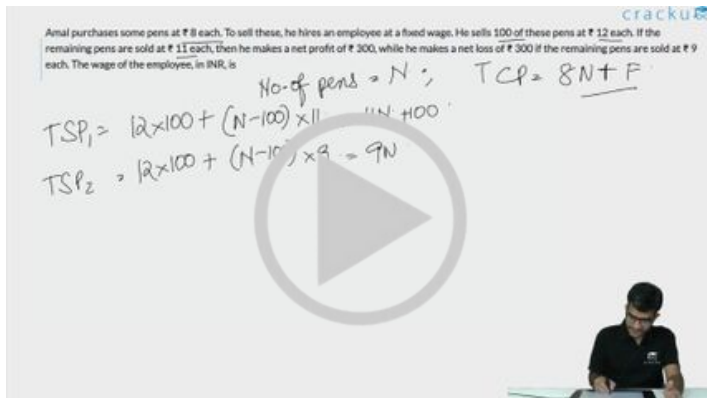
Amal purchases some pens at ₹ 8 each. To sell these, he hires an employee at a fixed wage. He sells 100 of these pens at ₹ 12 each. If the remaining pens are sold at ₹ 11 each, then he makes a net profit of ₹ 300, while he makes a net loss of ₹ 300 if the remaining pens are sold at ₹ 9 each. The wage of the employee, in INR, is

cracku

No. of pens = N ; $TCP = 8N + F$

$TSP_1 = 12 \times 100 + (N - 100) \times 11$

$TSP_2 = 12 \times 100 + (N - 100) \times 9$



VIDEO SOLUTION

Question 36

Raj invested ₹ 10000 in a fund. At the end of first year, he incurred a loss but his balance was more than ₹ 5000. This balance, when invested for another year, grew and the percentage of growth in the second year was five times the percentage of loss in the first year. If the gain of Raj from the initial investment over the two year period is 35%, then the percentage of loss in the first year is

- A 5
- B 15
- C 17
- D 10

Raj invested ₹ 10000 in a fund. At the end of first year, he incurred a loss but his balance was more than ₹ 5000. This balance, when invested for another year, grew and the percentage of growth in the second year was five times the percentage of loss in the first year. If the gain of Raj from the initial investment over the two year period is 35%, then the percentage of loss in the first year is

cracku

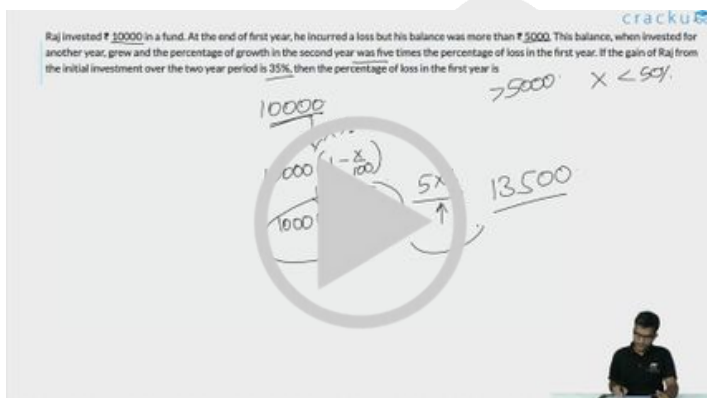
10000

$10000(1 - \frac{x}{100})$

$5x$

13500

$x < 50\%$



VIDEO SOLUTION

FREE

CAT Sectional Tests

cracku

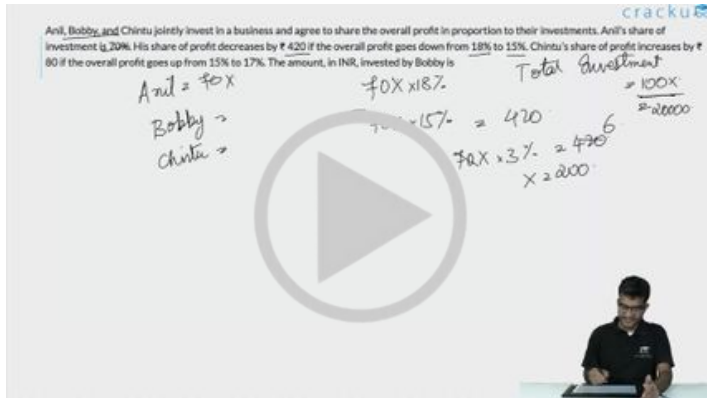
Take Here



Question 37

Anil, Bobby, and Chintu jointly invest in a business and agree to share the overall profit in proportion to their investments. Anil's share of investment is 70%. His share of profit decreases by ₹ 420 if the overall profit goes down from 18% to 15%. Chintu's share of profit increases by ₹ 80 if the overall profit goes up from 15% to 17%. The amount, in INR, invested by Bobby is

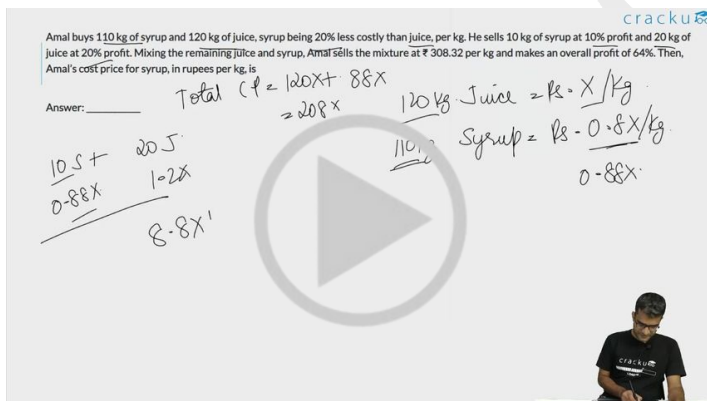
- A** 2000
- B** 2400
- C** 2200
- D** 1800



 VIDEO SOLUTION

Question 38

Amal buys 110 kg of syrup and 120 kg of juice, syrup being 20% less costly than juice, per kg. He sells 10 kg of syrup at 10% profit and 20 kg of juice at 20% profit. Mixing the remaining juice and syrup, Amal sells the mixture at ₹ 308.32 per kg and makes an overall profit of 64%. Then, Amal's cost price for syrup, in rupees per kg, is



VIDEO SOLUTION

Question 39

Gita sells two objects A and B at the same price such that she makes a profit of 20% on object A and a loss of 10% on object B. If she increases the selling price such that objects A and B are still sold at an equal price and a profit of 10% is made on object B, then the profit made on object A will be nearest to

- A 42%

- B 45%
- C 47%
- D 49%

cracku

Gita sells two objects A and B at the same price such that she makes a profit of 20% on object A and a loss of 10% on object B. If she increases the selling price such that objects A and B are still sold at an equal price and a profit of 10% is made on object B, then the profit made on object A will be nearest to

A) 42%
C) 47%

Handwritten notes:

A: $\frac{SP}{CP} = 1.2$
 $\frac{NSP}{CP} = \frac{1.2 \times 1.1}{0.9}$

B: $\frac{SP}{CP} = 0.9$
 $\frac{NSP}{CP} = 1$

Video solution showing a man at a desk.

VIDEO SOLUTION

cracku

CAT Study Material

All CAT Free Resources

Video solution showing two people at a desk.

Question 40

A merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him. He sells the same cloth at a rate of Rs.110 per meter but cheats his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers. If the merchant provides a 5% discount, the resulting profit earned by him is

- A 4.2%
- B 9.7%
- C 15.5%
- D 16%

cracku

A merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him. He sells the same cloth at a rate of Rs.110 per meter but cheats his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers. If the merchant provides a 5% discount, the resulting profit earned by him is

A) 4.2%
C) 15.5%

Handwritten notes:

Rs. 100 / m
 $0.05 \rightarrow 1.05$
 0.95

Video solution showing a man at a desk.

VIDEO SOLUTION

Question 41

Minu purchases a pair of sunglasses at Rs.1000 and sells to Kanu at 20% profit. Then, Kanu sells it back to Minu at 20% loss. Finally, Minu sells the same pair of sunglasses to Tanu. If the total profit made by Minu from all her transactions is Rs.500, then the percentage of profit made by Minu when she sold the pair of sunglasses to Tanu is

- A 35.42%
- B 52%
- C 31.25%
- D 26%

cracku

Minu purchases a pair of sunglasses at Rs.1000 and sells to Kanu at 20% profit. Then, Kanu sells it back to Minu at 20% loss. Finally, Minu sells the same pair of sunglasses to Tanu. If the total profit made by Minu from all her transactions is Rs.500, then the percentage of profit made by Minu when she sold the pair of sunglasses to Tanu is

A) 35.42%
B) 52%
C) 31.25%

Handwritten notes:
Minu: CP = Rs. 1000, SP = Rs. 1200, Profit = 200.
Kanu: CP = Rs. 1200, SP = Rs. 960, Loss = 240.
Minu: CP = Rs. 960, SP = Rs. 1260, Profit = 300.
Total Profit = 200 + 300 = 500.

Percentage of profit made by Minu when she sold the pair of sunglasses to Tanu is $\frac{300}{960} \times 100 = 31.25\%$.

VIDEO SOLUTION

Question 42

Jayant bought a certain number of white shirts at the rate of Rs 1000 per piece and a certain number of blue shirts at the rate of Rs 1125 per piece. For each shirt, he then set a fixed market price which was 25% higher than the average cost of all the shirts. He sold all the shirts at a discount of 10% and made a total profit of Rs.51000. If he bought both colors of shirts, then the maximum possible total number of shirts that he could have bought is

cracku

Jayant bought a certain number of white shirts at the rate of Rs 1000 per piece and a certain number of blue shirts at the rate of Rs 1125 per piece. For each shirt, he then set a fixed market price which was 25% higher than the average cost of all the shirts. He sold all the shirts at a discount of 10% and made a total profit of Rs.51000. If he bought both colors of shirts, then the maximum possible total number of shirts that he could have bought is

Answer: _____

Handwritten notes:
W → 1000, B → 1125.
Average cost = $\frac{1000W + 1125B}{W+B}$.
Market Price (MP) = $1.25 \times \text{Average cost}$.
Selling Price (SP) = $0.9 \times \text{MP}$.
Profit = $0.125 \times \frac{1000W + 1125B}{W+B} \times (W+B)$.
Total Profit = $0.125 \times (1000W + 1125B) = 51000$.
Solving for W and B, the maximum possible total number of shirts is 100.

VIDEO SOLUTION

cracku

CAT Daily Articles

Question 43

Bina incurs 19% loss when she sells a product at Rs. 4860 to Shyam, who in turn sells this product to Hari. If Bina would have sold this product to Shyam at the purchase price of Hari, she would have obtained 17% profit. Then, the profit, in rupees, made by Shyam is

cracku

Bina incurs 19% loss when she sells a product at Rs. 4860 to Shyam, who in turn sells this product to Hari. If Bina would have sold this product to Shyam at the purchase price of Hari, she would have obtained 17% profit. Then, the profit, in rupees, made by Shyam is

Answer: _____

Bina:
 $SP = 4860$
 $Loss = 19\%$
 $CP = \frac{4860}{1.19} = 4084.03$

Shyam:
 $CP = 4860$
 $SP = 4860$

Hari:
 $CP = 4860$
 $SP = 4860$

Profit made by Shyam = $SP - CP = 4860 - 4084.03 = 775.97$

VIDEO SOLUTION

Question 44

The selling price of a product is fixed to ensure 40% profit. If the product had cost 40% less and had been sold for 5 rupees less, then the resulting profit would have been 50%. The original selling price, in rupees, of the product is

- A 15
- B 14
- C 10
- D 20

cracku

The selling price of a product is fixed to ensure 40% profit. If the product had cost 40% less and had been sold for 5 rupees less, then the resulting profit would have been 50%. The original selling price, in rupees, of the product is

A) 15
 B) 14
 C) 10
 D) 20

Selling Price = $1.4X$
 Cost Price = X
 New CP = $0.6X$
 New SP = $1.4X - 5$

Profit = $SP - CP = 1.4X - 0.6X = 0.8X$

Profit = 50%
 $0.8X = 0.5(1.4X - 5)$
 $0.8X = 0.7X - 2.5$
 $0.1X = -2.5$
 $X = 25$

Original Selling Price = $1.4X = 1.4 \times 25 = 35$

VIDEO SOLUTION

Question 45

Gopi marks a price on a product in order to make 20% profit. Ravi gets 10% discount on this marked price, and thus saves Rs 15. Then, the profit, in rupees, made by Gopi by selling the product to Ravi, is

- A 10
- B 25
- C 15

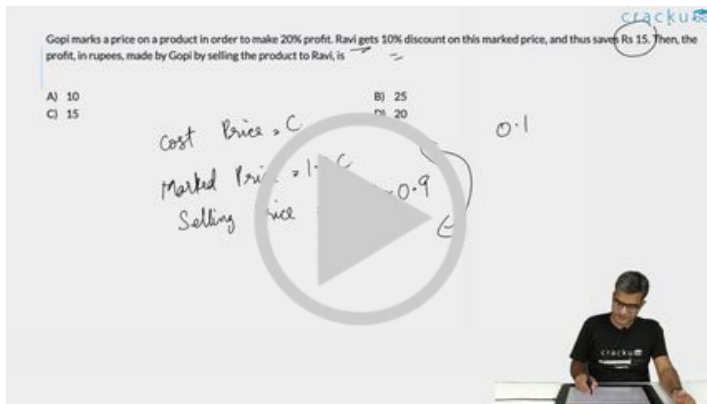
D 20

cracku

Gopi marks a price on a product in order to make 20% profit. Ravi gets 10% discount on this marked price, and thus saves Rs 15. Then, the profit, in rupees, made by Gopi by selling the product to Ravi, is

A) 10
B) 25
C) 15
D) 20

Cost Price = C
Marked Price = $1.2C$
Selling Price = $0.9 \times 1.2C = 1.08C$
Discount = 0.1



VIDEO SOLUTION

100+ CAT Quant Questions



Question 46

After two successive increments, Gopal's salary became 187.5% of his initial salary. If the percentage of salary increase in the second increment was twice of that in the first increment, then the percentage of salary increase in the first increment was

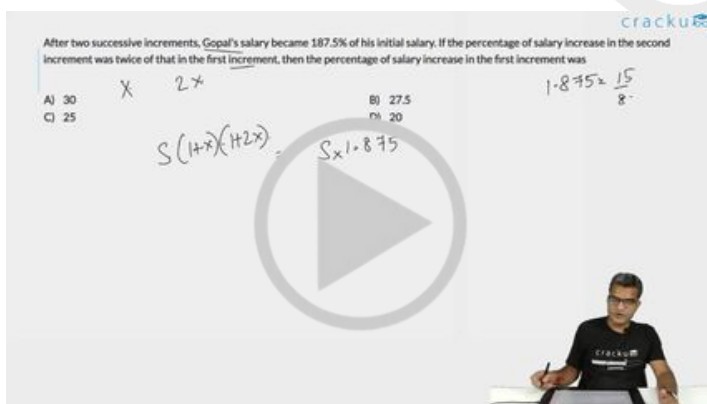
- A 30
- B 27.5
- C 25
- D 20

cracku

After two successive increments, Gopal's salary became 187.5% of his initial salary. If the percentage of salary increase in the second increment was twice of that in the first increment, then the percentage of salary increase in the first increment was

A) 30
B) 27.5
C) 25
D) 20

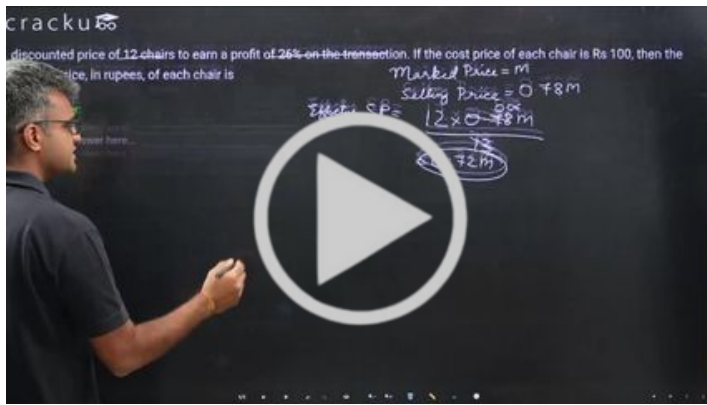
X $2X$
 $S(1+X)(1+2X) = S \times 1.875$
 $1.875 = \frac{15}{8}$



VIDEO SOLUTION

Question 47

A shopkeeper offers a discount of 22% on the marked price of each chair, and gives 13 chairs to a customer for the discounted price of 12 chairs to earn a profit of 26% on the transaction. If the cost price of each chair is Rs 100, then the marked price, in rupees, of each chair is



VIDEO SOLUTION

Question 48

The monthly sales of a product from January to April were 120, 135, 150 and 165 units, respectively. The cost price of the product was Rs. 240 per unit, and a fixed marked price was used for the product in all the four months. Discounts of 20%, 10% and 5% were given on the marked price per unit in January, February and March, respectively, while no discounts were given in April. If the total profit from January to April was Rs. 138825, then the marked price per unit, in rupees, was

- A 520
- B 525
- C 510
- D 515

VIEW SOLUTION

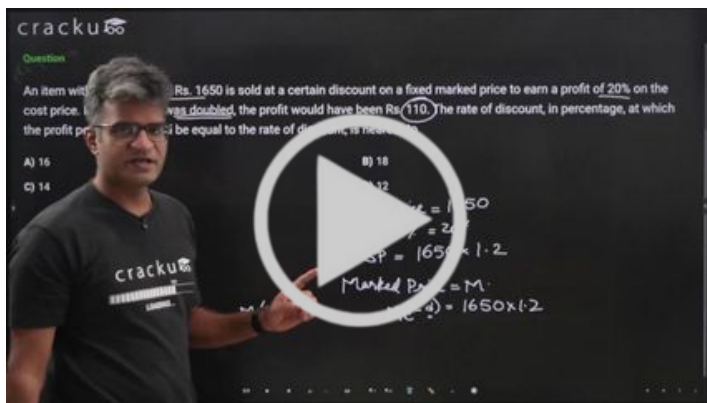
100+ CAT DILR Questions



Question 49

An item with a cost price of Rs. 1650 is sold at a certain discount on a fixed marked price to earn a profit of 20% on the cost price. If the discount was doubled, the profit would have been Rs. 110. The rate of discount, in percentage, at which the profit percentage would be equal to the rate of discount, is nearest to

- A 16
- B 18
- C 14
- D 12



VIDEO SOLUTION

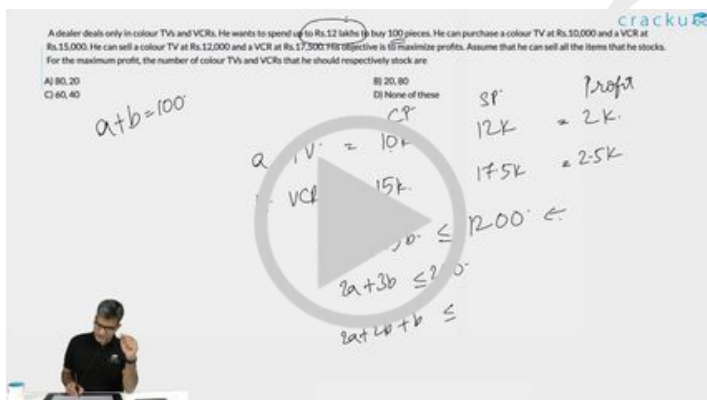
Instructions

A dealer deals only in colour TVs and VCRs. He wants to spend up to Rs.12 lakhs to buy 100 pieces. He can purchase a colour TV at Rs.10,000 and a VCR at Rs.15,000. He can sell a colour TV at Rs.12,000 and a VCR at Rs.17,500. His objective is to maximize profits. Assume that he can sell all the items that he stocks.

Question 50

For the maximum profit, the number of colour TVs and VCRs that he should respectively stock are

- A 80, 20
- B 20, 80
- C 60, 40
- D None of these



VIDEO SOLUTION

Question 51

If the dealer would have managed to get an additional space to stock 20 more items, then for maximizing profit, the ratio of number of VCRs and number of TVs that he should stock is

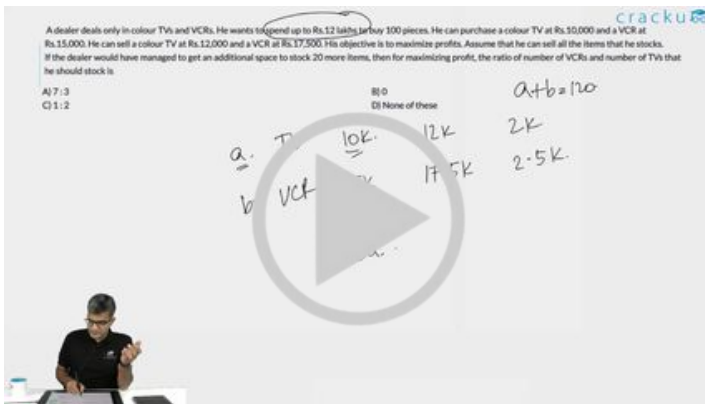
- A 7 : 3
- B 0
- C 1 : 2

D None of these

A dealer deals only in colour TVs and VCRs. He wants to spend up to Rs.12 lakh to buy 100 pieces. He can purchase a colour TV at Rs.15,000 and a VCR at Rs.11,000. He can sell a colour TV at Rs.12,000 and a VCR at Rs.17,500. His objective is to maximize profit. Assume that he can sell all the items that he stocks. If the dealer would have managed to get an additional space to stock 20 more items, then for maximizing profit, the ratio of number of VCRs and number of TVs that he should stock is

A) 7:3
B) 1:2
C) 1:2
D) None of these

Handwritten notes: $a+b=100$, $15a+11b \leq 12000$, $12a+17.5b$, $a+b=120$, $2K$, $2.5K$.



VIDEO SOLUTION

100+ CAT VARC Questions



Question 52

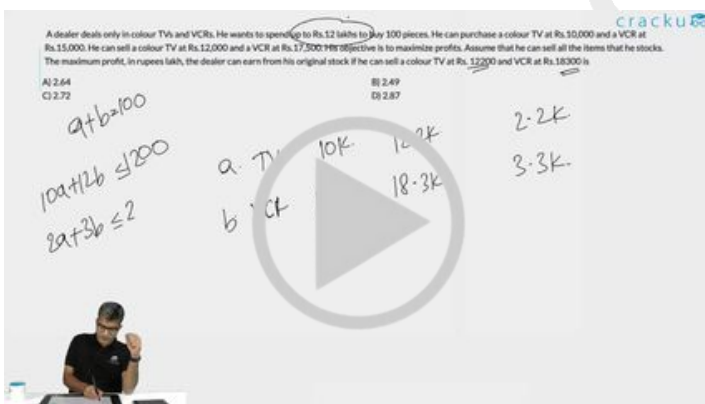
The maximum profit, in rupees lakh, the dealer can earn from his original stock if he can sell a colour TV at Rs. 12200 and VCR at Rs.18300 is

- A 2.64
- B 2.49
- C 2.72
- D 2.87

A dealer deals only in colour TVs and VCRs. He wants to spend up to Rs.12 lakh to buy 100 pieces. He can purchase a colour TV at Rs.15,000 and a VCR at Rs.11,000. He can sell a colour TV at Rs.12,000 and a VCR at Rs.17,500. His objective is to maximize profit. Assume that he can sell all the items that he stocks. The maximum profit, in rupees lakh, the dealer can earn from his original stock if he can sell a colour TV at Rs. 12200 and VCR at Rs.18300 is

A) 2.64
B) 2.72
C) 2.72
D) 2.87

Handwritten notes: $a+b=100$, $15a+11b \leq 12000$, $12a+18.3b$, $2.2K$, $3.3K$.



VIDEO SOLUTION

Instructions

Ghosh Babu has a certain amount of property consisting of cash, gold coins and silver bars. The cost of a gold coin is Rs. 4000 and the cost of a silver bar is Rs. 1000. Ghosh Babu distributed his property among his daughters equally. He gave to his eldest daughter gold coins worth 20% of the total property and Rs. 25000 in cash. The second daughter was given silver bars worth 20% of the remaining property and Rs. 50000 cash.

Among the third and fourth daughters, he distributed the remaining gold and silver bars equally both together accounting each for 20% of the property remaining after the previous distribution. He also gave the third and fourth daughters Rs. 25000 more than what the second daughter had received in cash.

Question 53

The amount of property in gold and silver possessed by Ghosh Babu is

- A 2,25,000
- B 2,75,000
- C Rs. 4,25,000
- D None of these

Ghosh Babu has a certain amount of property consisting of cash, gold coins and silver bars. The cost of a gold coin is Rs. 4000 and the cost of a silver bar is Rs. 1000. Ghosh Babu distributed his property among his daughters equally. He gave to his eldest daughter gold coins worth 20% of the total property and Rs. 25000 in cash. The second daughter was given silver bars worth 20% of the remaining property and Rs. 50000 cash. He then gave each of the third and fourth daughters equal number of gold coins and silver bars both together accounting each for 20% of the property remaining after the previous distribution and Rs. 25000 more than what the second daughter had received in cash.

$$\begin{array}{l} \text{G.} \quad \text{S.} \quad \text{Cash} \\ 4000G \quad 1000S \quad C \\ P = 4000G + 1000S + C \\ \text{Daughter 1} = 0.25P \\ = 1000G + 250S + \frac{C}{4} \\ 0.2 \times P + 25000 = 0.25P \\ 0.05P = 25000 \end{array}$$



VIDEO SOLUTION

Question 54

Total property of Ghosh Babu (in Rs.lakh) is

- A 5.0
- B 7.5
- C 10.0
- D 12.5.

Ghosh Babu has a certain amount of property consisting of cash, gold coins and silver bars. The cost of a gold coin is Rs. 4000 and the cost of a silver bar is Rs. 1000. Ghosh Babu distributed his property among his daughters equally. He gave to his eldest daughter gold coins worth 20% of the total property and Rs. 25000 in cash. The second daughter was given silver bars worth 20% of the remaining property and Rs. 50000 cash. He then gave each of the third and fourth daughters equal number of gold coins and silver bars both together accounting each for 20% of the property remaining after the previous distribution and Rs. 25000 more than what the second daughter had received in cash.

$$\begin{array}{l} \text{G.} \quad \text{S.} \quad \text{Cash} \\ 4000G \quad 1000S \quad C \\ P = 4000G + 1000S + C \\ \text{Daughter 1} = 0.25P \\ = 1000G + 250S + \frac{C}{4} \\ 0.2 \times P + 25000 = 0.25P \\ 0.05P = 25000 \end{array}$$



VIDEO SOLUTION

CAT Expected Questions PDF

cracku



Question 55

If the ratio of the gold to silver bars that Ghosh Babu had is 7:27, the number of silver bars he has is

- A 90
- B 60
- C 75
- D 135

Ghosh Babu has a certain amount of property consisting of cash, gold coins and silver bars. The cost of a gold coin is Rs. 4000 and the cost of a silver bar is Rs. 1000. Ghosh Babu distributed his property among his daughters equally. He gave to his eldest daughter gold coins worth 20% of the total property and Rs. 25000 in cash. The second daughter was given silver bars worth 20% of the remaining property and Rs. 50000 cash. He then gave each of the third and fourth daughters equal number of gold coins and silver bars both together accounting each for 20% of the property remaining after the previous distribution and Rs. 25000 more than what the second daughter had received in cash.

$$\begin{aligned} &G: S: \text{Cash} \\ &4000G: 1000S: C \\ &P = 4000G + 1000S + C \\ &\text{Daughter} = \frac{0.25P}{4} \\ &= \frac{1000G + 250S + \frac{C}{4}}{4} \\ &0.2 \times P + 25000 = 0.25P \\ &0.05P = 25000 \end{aligned}$$

VIDEO SOLUTION

Instructions

Answer the questions based on the following information. A watch dealer incurs an expense of Rs. 150 for producing every watch. He also incurs an additional expenditure of Rs. 30,000, which is independent of the number of watches produced. If he is able to sell a watch during the season, he sells it for Rs. 250. If he fails to do so, he has to sell each watch for Rs. 100.

Question 56

If he is able to sell only 1,200 out of 1,500 watches he has made in the season (and the rest 300 are sold out of season), then he has made a profit of

- A Rs. 90,000
- B Rs. 75,000
- C Rs. 45,000
- D Rs. 60,000

Answer the questions based on the following information: A watch dealer incurs an expense of Rs. 150 for producing every watch. He also incurs an additional expenditure of Rs. 30,000 which is independent of the number of watches produced. If he is able to sell a watch during the season, he sells it for Rs. 250. If he fails to do so, he has to sell each watch for Rs. 100.

If he is able to sell only 1,200 out of 1,500 watches he has made in the season (and the rest 300 are sold out of season), then he has made a profit of

A) Rs. 90,000
B) Rs. 70,000
C) Rs. 45,000
D) Rs. 60,000

$TCP = 150 \times 1500 + 30000 = 1000[225 + 30]$
 $TSP = 1200 \times 250 + 300 \times 100 = 1000[300 + 30]$
 $Profit = TSP - TCP = 1000[300 + 30 - 225 - 30] = 1000 \times 75 = 75,000$

VIDEO SOLUTION

Question 57

If he produces 1,500 watches, what is the number of watches that he must sell during the season in order to break-even, given that he is able to sell all the watches produced?

- A 500
- B 700
- C 800
- D 1,000

Answer the questions based on the following information: A watch dealer incurs an expense of Rs. 150 for producing every watch. He also incurs an additional expenditure of Rs. 30,000 which is independent of the number of watches produced. If he is able to sell a watch during the season, he sells it for Rs. 250. If he fails to do so, he has to sell each watch for Rs. 100.

If he is able to sell only 1,200 out of 1,500 watches he has made in the season (and the rest 300 are sold out of season), then he has made a profit of

A) Rs. 90,000
B) Rs. 70,000
C) Rs. 45,000
D) Rs. 60,000

$TCP = 150 \times 1500 + 30000 = 1000[225 + 30]$
 $TSP = 1200 \times 250 + 300 \times 100 = 1000[300 + 30]$
 $Profit = TSP - TCP = 1000[300 + 30 - 225 - 30] = 1000 \times 75 = 75,000$

VIDEO SOLUTION

CAT Important Topics

Instructions

Answer the following questions based on the following information.

A company purchases components A and B from Germany and USA respectively. A and B form 30% and 50% of the total production cost. Current gain is 20%. Due to change in the international scenario, cost of the German mark increased by 30% and that of USA dollar increased by 22%. Due to market conditions, the selling price cannot be increased by more than 10%.

Question 58

What is the maximum current gain possible?

- A 10%
- B 12.5%
- C 0%
- D 7.5%

 VIEW SOLUTION

Question 59

If the USA dollar becomes cheap by 12% over its original cost and the cost of German mark increased by 20%, what will be the gain? (The selling price is not altered.)

- A 10%
- B 20%
- C 15%
- D 7.5%

 VIEW SOLUTION

Instructions

DIRECTIONS for questions:

These questions are based on the situation given below:

Recently, Ghosh Babu spent his winter vacation on Kyakya Island. During the vacation, he visited the local casino where he came across a new card game. Two players, using a normal deck of 52 playing cards, play this game. One player is called the Dealer and the other is called the Player. First, the Player picks a card at random from the deck. This is called the base card. The amount in rupees equal to the face value of the base card is called the base amount. The face values of Ace, King, Queen and Jack are ten. For other cards, the face value is the number on the card. Once, the Player picks a card from the deck, the Dealer pays him the base amount. Then the dealer picks a card from the deck and this card is called the top card. If the top card is of the same suit as the base card, the Player pays twice the base amount to the Dealer. If the top card is of the same colour as the base card (but not the same suit) then the Player pays the base amount to the Dealer. If the top card happens to be of a different colour than the base card, the Dealer pays the base amount to the Player. Ghosh Babu played the game 4 times. First time he picked eight of clubs and the Dealer picked queen of clubs. Second time, he picked ten of hearts and the dealer picked two of spades. Next time, Ghosh Babu picked six of diamonds and the dealer picked ace of hearts. Lastly, he picked eight of spades and the dealer picked jack of spades. Answer the following questions based on these four games.

Question 60

If Ghosh Babu stopped playing the game when his gain would be maximized, the gain in Rs. would have been

- A 12
- B 20
- C 16
- D 4

 [VIEW SOLUTION](#)



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Question 61

The initial money Ghosh Babu had (before the beginning of the game sessions) was Rs. X . At no point did he have to borrow any money. What is the minimum possible value of X ?

- A 16
- B 8
- C 100
- D 24

 [VIEW SOLUTION](#)

Question 62

If the final amount of money that Ghosh Babu had with him was Rs. 100, what was the initial amount he had with him?

- A 120
- B 8
- C 4
- D 96

 [VIEW SOLUTION](#)

Answers

1.C	2.B	3.B	4.D	5.D	6.C	7.A	8.C
9.A	10.C	11.A	12.C	13.B	14.A	15.D	16.B
17.A	18.70000	19.D	20.A	21.B	22.D	23.B	24.A
25.A	26.C	27.A	28.C	29.C	30.20000	31.D	32.D
33.C	34.A	35.1000	36.D	37.A	38.160	39.C	40.C
41.C	42.407	43.2160	44.B	45.A	46.C	47.175	48.B
49.C	50.C	51.B	52.A	53.B	54.A	55.D	56.B
57.B	58.A	59.B	60.A	61.B	62.D		

Explanations

1. C

Let a, b and c be the percentages of amount invested in options A, B and C respectively $\Rightarrow a + b + c = 100$

Return attained if there is a rise in the stock market $\Rightarrow 0.001a + 0.05b - 0.025c$

Return attained if there is a fall in the stock market $\Rightarrow 0.001a - 0.03b + 0.02c$

Maximum guaranteed return is attained when both are equal because it is indifferent to rise and fall in the market.

$$0.001a + 0.05b - 0.025c = 0.001a - 0.03b + 0.02c$$

$$\Rightarrow 0.08b = 0.045c \Rightarrow 16b = 9c$$

Let's put the values for a, b and c that satisfy the above equation.

$$b = 9, c = 16, a = 75 \Rightarrow \text{return} = 0.125$$

$$b = 18, c = 32, a = 50 \Rightarrow \text{return} = 0.15$$

$$b = 27, c = 48, a = 25 \Rightarrow \text{return} = 0.175$$

$$b = 36, c = 64, a = 0 \Rightarrow \text{return} = 0.2$$

Hence, the maximum guaranteed return is 0.2%

 VIDEO SOLUTION

2. B

Let a, b and c be the percentages of amount invested in options A, B and C respectively $\Rightarrow a + b + c = 100$

Return attained if there is a rise in the stock market $\Rightarrow 0.001a + 0.05b - 0.025c$

Return attained if there is a fall in the stock market $\Rightarrow 0.001a - 0.03b + 0.02c$

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$$b = 27, c = 48, a = 25 \Rightarrow \text{return} = 0.175$$

$$b = 36, c = 64, a = 0 \Rightarrow \text{return} = 0.2$$

Hence, the maximum guaranteed return is 0.2% and it is attained when 36% is invested in option B and 64% is invested in option C.

 VIDEO SOLUTION

3. B

$$\text{Cost of 20 units} = 240 + 20b + 400c$$

$$\text{Cost of 40 units} = 240 + 40b + 1600c = \frac{5}{3} * (240 + 20b + 400c) \Rightarrow 720 + 120b + 4800c = 1200 + 100b + 2000c$$

$$\Rightarrow 480 = 20b + 2800c \Rightarrow 120 = 5b + 700c$$

$$\text{Cost of 60 units} = 240 + 60b + 3600c = \frac{3}{2} (240 + 40b + 1600c) \Rightarrow 480 + 120b + 7200c = 720 + 120b + 4800c$$

$$\Rightarrow 240 = 2400c \Rightarrow c = 1/10 \text{ and } b = 10$$

Let the number of items needed for max profit be k

$$CP = 240 + 10k + \frac{k^2}{10}$$

$$SP = 30k$$

$$\text{Profit} = SP - CP = 30k - 240 - 10k - \frac{k^2}{10} = 20k - 240 - \frac{k^2}{10}$$

$$\text{or Profit} = \frac{1}{10}(-k^2 + 200k - 2400)$$

$$\text{or, Profit} = \frac{1}{10}(-(k^2 - 200k + 2400))$$

$$\text{or, Profit} = \frac{1}{10}(-(k^2 - 200k + 2400 + 7600 - 7600))$$

$$\text{or, Profit} = \frac{1}{10}(-(k^2 - 200k + 10000) + 7600)$$

$$\text{or, Profit} = \frac{1}{10}(-(k - 100)^2 + 7600)$$

To maximise the value of Profit, $-(k - 100)^2$ must be 0.

So, k must be equal to 100.

Hence, option B is the correct answer.

 VIDEO SOLUTION

4. D

$$\text{Cost of 20 units} = 240 + 20b + 400c$$

$$\text{Cost of 40 units} = 240 + 40b + 1600c = \frac{5}{3} * (240 + 20b + 400c) \Rightarrow 720 + 120b + 4800c = 1200 + 100b + 2000c$$

$$\Rightarrow 480 = 20b + 2800c \Rightarrow 120 = 5b + 700c$$

$$\text{Cost of 60 units} = 240 + 60b + 3600c = \frac{3}{2} (240 + 40b + 1600c) \Rightarrow 480 + 120b + 7200c = 720 + 120b + 4800c$$

$$\Rightarrow 240 = 2400c \Rightarrow c = 1/10 \text{ and } b = 10$$

Let the number of items needed for max profit be k

$$CP = 240 + 10k + \frac{k^2}{10}$$

$$SP = 30k$$

$$\text{Profit} = SP - CP = 30k - 240 - 10k - \frac{k^2}{10} = 20k - 240 - \frac{k^2}{10}$$

$$\text{Maximum when } 20 - \frac{k}{5} = 0 \text{ or } k = 100$$

$$\text{Profit} = 2000 - 240 - 1000 = 760$$

 VIDEO SOLUTION

5. D

In total Raju bets 6000Rs and ends up with no profit - no loss. So there are 3 possibilities.

- 1) White comes 2nd, Black comes 4th and Red comes 5th.
- 2) Black comes 1st, White comes 3rd and Red comes 4th or 5th.
- 3) Black comes 2nd, Red comes 3rd and White comes 4th or 5th.

So there can never be 3 horses between white and red according to above to possibilities. Hence option D cannot be true.

 VIDEO SOLUTION



6. C

There are total 3 cases which satisfies the condition "no profit and no loss."

Case 1: White comes 2nd.(remaining two horses(red/black) come 4th/5th)

Profit from white horse = Final Amount - Initial Amount = $2000 \times 3 - 2000 = 4000$

Loss from Red and Black horse = $3000 + 1000 = 4000$

Net profit = $4000 - 4000 = 0$

Case 2: Black, Red come second, third respectively.(remaining one horse(white) comes 4th/5th)

Profit from Black = $1000 \times 3 - 1000 = 2000$

Profit from Red = $3000 - 3000 = 0$

Loss from white = 2000

Net profit = $2000 - 2000 = 0$

Case 3: black, white come first, third respectively.(remaining one horse(red) comes 4th/5th)

Profit from Black = $1000 \times 4 - 1000 = 3000$

Profit from White = $2000 - 2000 = 0$

Loss from Red = 3000

Net Profit = $3000 - 3000 = 0$

And it is mentioned that grey case 4th. $\Rightarrow$ case 1 is wrong.(because, in that case red, black should come 4th,5th)

So option C cannot be true.

 VIDEO SOLUTION

7. A

Let the price of the painting be P

One cycle of price increase and decrease reduces the price by $x^2/100 * P = 441$

Let the new price be N $\Rightarrow P - x^2/100 * P = N$

Price after the second cycle = $N - x^2/100 * N = 1944.81$

$\Rightarrow (P - x^2/100 * P)(1 - x^2/100) = 1944.81$

$\Rightarrow (P - 441)(1 - 441/P) = 1944.81$

$\Rightarrow P - 441 - 441 + 441^2/P = 1944.81$

$\Rightarrow P^2 - (882 + 1944.81)P + 441^2 = 0$

$$\Rightarrow P^2 - 2826.81P + 441^2 = 0$$

From the options, the value 2756.25 satisfies the equation.

So, the price of the article is Rs 2756.25

 VIDEO SOLUTION

8. C

Let's say total money was x rs.

So cost price of 40 mango will be $= x$;

Hence cost price of 20 mangoes will be $= \frac{x}{2}$

Taxi fare $= \frac{10x}{100}$

Total expense $= \frac{x}{2} + \frac{10x}{100} = \frac{6x}{10}$

Remaining money $= \frac{4x}{10}$

Cost price of 1 orange will be $= \frac{x}{50}$

Hence in $\frac{4x}{10}$ rs. 20 oranges can be purchased.

 VIDEO SOLUTION

9. A

Let's say the cost of the cloth is x rs per metre. Because of the faulty meter, he is paying x for 120 cms when buying.

So cost of 100 cms $= 100x/120$.

He is selling 80 cms for x , so selling price of 100cms of cloth is $100x/80$.

discount $= 20\%$

so the effective selling price is $.8 * 100x/80 = x$

profit $= SP - CP = x - 100x/120 = x/6$

Profit % $= x/6$ divided by $100x/120 = 20\%$

 VIDEO SOLUTION

10. C

Let the original weight of the diamond be equal to $10k$. So, after breaking into 4 pieces, the parts of the diamond weight $k, 2k, 3k, 4k$

The price of the diamond varies directly in proportion to the weight. Let us assume, the $P = C * W^2$ where C is a constant and W is the weight of the diamond.

Therefore, the original price is $C * 10k * 10k = 100k^2 * C$

The new weight is $Ck^2 + C(2k)^2 + C(3k)^2 + C(4k)^2 = 30k^2C$

The decrease in the price equals 70,000. So, $100k^2C - 30k^2C = 70000$

Or, $k^2C = 1000$

Therefore the original price $= 100k^2C = 100000$

 VIDEO SOLUTION



11. A

As shopkeeper gave 3 one-rupee change for 20 rs. change, Buyer must have ordered for a total of 17 rs. stamps.

Now buyer ordered for at least more than 1 stamp for each type

Hence the minimum he bought was:

2 stamp for 5 rupees = 10 rs.

2 stamp for 2 rupees = 4 rs.

2 stamp for 1 rupee = 2 rs.

For the total to be seventeen, the buyer must have purchased 3 one rupee stamps.

And 3 one rupee stamps were also there as changes given by shopkeeper.

So total number of stamps = $2+2+(3+3) = 10$

[VIDEO SOLUTION](#)

12. C

let's say price of maruti car is x rs.

Sales = y

revenue = xy

Changed price = 1.3x

changed value of sales = 0.8y

new revenue = 1.04 xy

Percentage change in revenue = 4%

[VIEW SOLUTION](#)

13. B

Selling price of first watch = 300

Profit = 10%

cost price = $\frac{300}{1.1}$

Selling price of second watch = 300

Loss = 10%

cost price = $\frac{300}{0.9}$

Total selling price of transaction = 600

Total cost price of transaction = $300\left(\frac{10}{11} + \frac{10}{9}\right) = 600\left(\frac{100}{99}\right)$

Loss = $600\left(\frac{100}{99} - 1\right)$

%loss = $\left(600\left(\frac{100}{99} - 1\right)\right) \div \left(600\left(\frac{100}{99}\right)\right) \times 100 = 1$

[VIDEO SOLUTION](#)

14. A

Let's say cost price is 100

gain = 14.28

selling price = 114.28

Marked price = x(say)

So $x - \frac{11.11x}{100} = \frac{8x}{9} = 114.28$

Or $x = 128.52$

So marked price is 28.52% more than cost price.

 VIDEO SOLUTION

15. D

Let's say he buy fruits of weights 3 kg., 4kg., 5 kg.

Total kilograms of dry fruits = $3 + 4 + 5 = 12$

Overall cost price = $3 \cdot 100 + 4 \cdot 80 + 5 \cdot 60 = 300 + 320 + 300 = 920$

So cost price per kg. = $\frac{300 + 320 + 300}{12} = \frac{920}{12}$

Selling price = $\frac{920}{12} \times \frac{3}{2} = 115$ per kg (Since Profit is 50%)

Hence answer will be D.

 VIDEO SOLUTION



16. B

He earns $x\%$ on first 2000 and $y\%$ on rest of his income.

So on 4000 rs. , he will earn as follows:

$$2000 \frac{x}{100} + 2000 \frac{y}{100} = 700$$

$$\text{Or } x + y = 35$$

Similarly on 5000 rs. ,he will earn 900 as follows:

$$2000 \frac{x}{100} + 3000 \frac{y}{100} = 900$$

$$\text{Or } 20x + 30y = 900$$

On solving above equations, we will get value of $x = 15$

 VIDEO SOLUTION

17. A

Let's say price of turban is x .

So total price for 12 months will be = $90 + x$

$$\text{total price for 9 months} = \frac{(90+x) \times 9}{12} = (65 + x)$$

By solving above equation, we will get value of $x = 10$.

 VIDEO SOLUTION

18. 70000

Let his total savings be $100x$.

He invests $50x$ in fixed deposits. 30% of $50x$, which is $15x$ is invested in stocks and $35x$ goes to savings bank.

It is given $85x = 59500$

$$x = 700$$

Hence, $100x = 70000$

 VIDEO SOLUTION

19. **D**

Let the retail price be M and cost price be C .

Given,

$$0.85 M = 1.02 C$$

$$M = 1.2 C$$

If he wants 20% profit he has to sell at $1.2C$, which is nothing but the retail price.

 VIDEO SOLUTION

20. **A**

Given,

$$C1 : C2 : C3 = 9 : 10 : 8 \dots i$$

$$C2 : C4 : C5 = 18 : 19 : 20 \dots ii$$

Let's multiply i by 9 and ii by 5

$$C1 : C2 : C3 = 81 : 90 : 72$$

$$C2 : C4 : C5 = 90 : 95 : 100$$

$$\text{Therefore, } C1 : C2 : C3 : C4 : C5 = 81 : 90 : 72 : 95 : 100$$

Given,

$$100x - 81x = 19$$

$$x = 1$$

$$\text{Hence, total profit} = 100 + 95 + 72 + 90 + 81 = 438$$

 VIDEO SOLUTION



CAT Syllabus PDF



21. **B**

Let the cost of good mangoes be $2x$ per kg. The cost of medium mangoes be x per kg.

$$\text{CP of good mangoes} = 160x$$

$$\text{CP of medium mangoes} = 40x$$

$$\text{His selling price} = 0.9 \times 2x = 1.80x$$

$$\text{Therefore, total revenue generated by selling all the mangoes} = 120 \times 1.8x = 216x$$

$$\text{Hence, the profit \%} = \frac{16x}{200x} \times 100 = 8\%$$

 VIDEO SOLUTION

22. **D**

Let the cost price be C and the marked price be M .

Given,

$$0.6 M = 1.2 C$$

$$M = 2C$$

$$\text{CP of 60 toys} = 60C$$

Now only 50 are remaining.

Hence,

$$M(1 - d) \times 50 = 72C$$

$$1 - d = 0.72$$

$$d = .28$$

Hence 28%

 VIDEO SOLUTION

23. **B**

Let the manufacturing price of the table = x

Hence the price at which the wholesaler bought from the manufacturer = $1.1 \times x$

The price at which the retailer bought from the wholesaler = $1.3 \times 1.1 \times x$

The price at which the customer bought from the retailer = $1.5 \times 1.3 \times 1.1 \times x$

$$1.5 \times 1.3 \times 1.1 \times x = 4290$$

$$\Rightarrow x = 2000$$

 VIDEO SOLUTION

24. **A**

Let the number of dozens of candies he bought of each variety be x

$$\text{Hence total cost} = 12x + 15x = 27x$$

$$\text{Total selling price} = 16.50 \times 2x = 33x$$

$$\text{Profit} = 33x - 27x = 6x$$

$$\text{Given } 6x = 150 \Rightarrow x = 25$$

Hence he bought 50 dozens of candies in total

 VIDEO SOLUTION

25. **A**

CP of the table at which the shopkeeper procured each table = p

It is given that shopkeeper sold the tables to Amal and Asim at a profit of 20% and at a loss of 20%, respectively

The selling price of the tables = $1.2p$ and $0.8p$ to Amal and Asim respectively.

Amal sells his table to Bimal at a profit of 30%

$$\text{So, CP of the table by Bimal (x)} = 1.2p \times 1.3 = 1.56p$$

Asim sells his table to Barun at a loss of 30%

$$\text{So, CP of the table by Barun (y)} = 0.7 \times 0.8p = 0.56p$$

$$(x - y)/p = (1.56p - 0.56p)/p = p/p = 1$$

 VIDEO SOLUTION



26. C

Let the cost of each bicycle = $100b$

CP of 10 bicycles = $1000b$

It is given that he sold six of these at a profit of 25% and the remaining four at a loss of 25%

SP of 10 bicycles = $125b \times 6 + 75b \times 4$

= $1050b$

Profit = $1050b - 1000b = 50b$

$50b = 2000$

CP = $100b = 4000$

 VIDEO SOLUTION

27. A

Assuming the income of Bimla = $100a$, then the income of Amala will be $120a$.

And the income of Kamala will be $120a \times 100/80 = 150a$

If Kamala's income goes down by 4%, then new income of Kamala = $150a - 150a(4/100) = 150a - 6a = 144a$

If Bimla's income goes up by 10 percent, her new income will be $100a + 100a(10/100) = 110a$

=> Hence the Kamala income will exceed Bimla income by $(144a - 110a) \times 100/110a = 31$

 VIDEO SOLUTION

28. C

Assuming the maximum marks = $100a$, then Meena got $40a$

After increasing her score by 50%, she will get $40a(1 + 50/100) = 60a$

Passing score = $60a + 35$

Post review score after 20% increase = $60a \times 1.2 = 72a$

=> Hence, $60a + 35 + 7 = 72a$

=> $12a = 42$ => $a = 3.5$

=> maximum marks = 350 and passing marks = $210 + 35 = 245$

=> Passing percentage = $245 \times 100/350 = 70$

 VIDEO SOLUTION

29. C

Assuming the cost price of pen = $100p$ and the cost price of book = $100b$

So, on selling a pen at 5% loss and a book at 15% gain, net gain = $-5p + 15b = 7$ 1

On selling the pen at 5% gain and the book at 10% gain, net gain = $5p + 10b = 13$ 2

Adding 1 and 2 we get, $25b = 20$

Hence $100b = 20 \times 4 = 80$,

C is the answer.

 VIDEO SOLUTION

30. **20000**

Let the price of desktop and laptop be x, y respectively.

Given,

$$x + y = 50000 \dots (i)$$

$$1.2x + 0.9y = 50000(1.02) = 51000 \dots (ii)$$

$(ii) - 0.9(i)$ gives

$$0.3x = 6000 \Rightarrow x = 20000.$$

 VIDEO SOLUTION



31. **D**

Let ' x ' be the strength of group G. Based on the information, $0.65x$ constitutes of literate people {the rest $0.35x$ = illiterate}

Of this $0.65x$, 75% are old people $= (0.75)0.65x$ old literates. The total number of old people in group G is $0.72x$ {72% of the total}. Thus, the total number of old people who are illiterate $= 0.72x - 0.4875x = 0.2325x$.

This is $\frac{0.2325x}{0.35x} \times 100 \approx 66\%$ of the total number of illiterates. Hence, Option D is the correct answer.

 VIDEO SOLUTION

32. **D**

Let the total marks be $100x$

Marks obtained by Bishnu = $52x$

Marks obtained by Asha = $64x$

Marks obtained by Ramesh = $52x + 23$

Marks obtained by Ramesh = $64x - 34$

$$\Rightarrow 52x + 23 = 64x - 34$$

$$\Rightarrow x = \frac{19}{4}$$

Marks obtained by Geeta = $84x = 84 \cdot 19/4 = 399$

 VIDEO SOLUTION

33. **C**

Let the cost price of 1kg of sugar = Rs 100

The total cost price of 35 kg = Rs 3500

Marked up price per kg = Rs 120

Given, the final profit is 15% $\Rightarrow$ Final SP of 35 kg = $3500 \cdot 1.15 = \text{Rs } 4025$

First 5 kg's are sold at 20% marked up price $\Rightarrow SP_1 = 5 \cdot 100 \cdot 1.2 = \text{Rs } 600$

Next 15 kgs are sold after giving 10% discount $\Rightarrow SP_2 = 15 \cdot 100 \cdot 1.2 \cdot 0.9 = 1620$

3kgs of sugar got wasted

=> 23 kg of sugar was sold at Rs (600 +1620) = Rs 2220

Remaining 12kg should be sold at Rs 4025 - 2220 = Rs1805

=> SP of 1kg = 1805/12 $\simeq$ 150

Hence, the seller should further mark up by $\frac{(150-120)}{120} \cdot 100 = 25\%$

 VIDEO SOLUTION

34. A

Let the CP of the each toy be "x". CP of 12 toys will be "12x". Now the shopkeeper made a 10% profit on CP. This means that

$12x(1.1) = 2112$ or $x=160$. Hence the CP of each toy is ₹160.

Now let the SP of each toy be "m". Now he sold 8 toys at 20% discount. This means that $8m(0.8)$ or $6.4m$

He sold 4 toys at an additional 25% discount. $4m(0.8)(0.75)=2.4m$ Now $6.4m+2.4m=8.8m=2112$ or $m=240$

Hence CP= 160 and SP=240. Hence profit percentage is 50%.

 VIDEO SOLUTION

35. 1000

Let the number of pens purchased be n. Then the cost price is 8n. The total expenses incurred would be $8n+W$, where W refers to the wage.

Then SP in the first case = $12 \times 100 + 11 \times (n - 100)$

Given profit is 300 in this case: $1200+11n-1100-8n-W=300 \Rightarrow 3n-W = 200$

In second case: $1200+9n-900-8n-W=-300$ (Loss). $\Rightarrow W-n = 600$.

Adding the two equations: $2n = 800$

$n = 400$.

Thus $W = 600 + 400 = 1000$

 VIDEO SOLUTION

CAT Percentile Predictor



36. D

Raj invested Rs 10000 in the first year. Assuming the loss he faced was x%.

The amount after 1 year is $10,000 \cdot (1 - x/100) = 10000 - 100x$.

Given the balance was greater than Rs 5000 and hence $x < 50$ percent.

When Raj invested this amount in the second year he earned a profit which is five times that of the first-year percentage.

Hence the amount after the second year is : $(10000 - 100x)(1 + \frac{5 \cdot x}{100})$.

Raj gained a total of 35 percent over the period of two years and hence the 35 percent is Rs 3500.

Hence the final amount is Rs 13,500.

$(10000 - 100x)(1 + \frac{5 \cdot x}{100}) = 13,500$

$(100 + 5 \cdot x) \cdot (100 - x) = 13500$

$10000 - 100x + 500x - 5x^2 = 13500$.

$$5x^2 - 400x + 3500 = 0$$

Solving the equation the roots are :

$$x = 10, x = 70.$$

Since $x < 50$, $x = 10$ percent.

 VIDEO SOLUTION

37. A

Let the amount invested by Anil Bobby and Chintu be x , y , and z .

Considering $x+y+z = 100*p$.

Given Anil's share was 70 percent = $70*p$.

As per the information provided :

His share of profit decreases by ₹ 420 if the overall profit goes down from 18% to 15%.

Since the profits are distributed in the ratio of their investments :

With a 3% decrease in the profits the value of profit earned by A decreased by Rs 420 which was 70 percent of the total invested.

$$\text{Hence for all three of them would be combinedly losing } (420) \cdot \left(\frac{10}{7}\right) = 600$$

Hence 3 percent profit was equivalent to Rs 600.

The initial investment is equivalent to Rs 20000.

This is the total amount invested.

Chintu's profit share increased by Rs 80 when the profit percentage increased by 2 %. A 2 percent increase in profit is equivalent to $\text{Rs } 20000 \cdot 2/100 = \text{Rs } 400$.

Of which Rs 80 is earned by Chintu which is 20% of the total Rs 400.

Hence he invested 20% of the total amount.

Bobby invested the other 10 percent.

$$10 \text{ percent of Rs } 20000 = \text{Rs } 2000$$

 VIDEO SOLUTION

38. 160

Total syrup - 110 kg

Total juice - 120 kg

It is given, cost price of syrup is 20% less than the cost price of juice.

Let the cost price of juice per kg be 10CP

Cost price of syrup per kg is 8CP

10kg syrup -> cost price = 80CP

It is given, 10kg syrup is sold at 10% profit. This implies selling price = $1.1 \cdot 80\text{CP} = 88\text{CP}$

20kg juice -> cost price = 200CP

It is given, 20kg juice is sold at 20% profit. This implies selling price = $1.2 \cdot 200\text{CP} = 240\text{CP}$

It is given, Mixing the remaining juice and syrup, Amal sells the mixture at ₹ 308.32 per kg

$$\text{Selling price of the remaining mixture} = 308.32 \cdot 200 = \text{Rs } 61664$$

$$\text{Total S.P} = 61664 + 328\text{CP}$$

$$\text{Total C.P} = 880\text{CP} + 1200\text{CP} = 2080\text{CP}$$

Overall profit = 64%

$$61664 + 328CP = \frac{164}{100} (2080CP)$$

Solving, we get CP = 20

Cost price for syrup per kg = 8CP = 8*20 = Rs 160

[VIDEO SOLUTION](#)

39. C

Let us assume the initial selling prices of A and B is p.

Given, she made profit of 20% on A $\Rightarrow 1.2 * c = p \Rightarrow c = 5p/6 \Rightarrow$ cost of A is $\frac{5}{6}p$

Given, she made a loss of 10% on B $\Rightarrow 0.9 * c = p \Rightarrow c = 10p/9 \Rightarrow$ cost of B is $\frac{10}{9}p$

Now, she sold them at a price such that a 10% profit is made on B

$$\Rightarrow \text{Selling price} = s = 11/10 * 10/9 p \Rightarrow \frac{11}{9}p$$

$$\Rightarrow \text{Profit \% on A} = \frac{\left(\frac{11}{9} - \frac{5}{6}\right)}{\left(\frac{5}{6}\right)} \times 100 = 46.66\% = \text{nearly } 47\%$$

[VIDEO SOLUTION](#)

40. C

It is given that a merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him.

Hence, the cost price of 105 cm clothes is 100 rupees.

It is also known that he marked the price of 100 cm clothes as 110 rupees, and gave a 5% discount, and he cheated his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers.

Hence, the selling price of 95 cm clothes is $110 * (19/20)$ rupees.

Therefore, the selling price of 105 cm clothes is 115.5 rupees.

Hence, the profit is 15.5%

The correct option is C

[VIDEO SOLUTION](#)



CAT Score Calculator



41. C

The cost price of the sunglass for Meenu when he purchased it for the first time was 1000 rupees, and he sold it to Kanu at 20% profit. Hence, the selling price of the sunglass is 1200 rupees, which Kanu purchased. Hence, the profit made by Meenu is $(1200 - 1000) = 200$ rupees.

Hence, the cost price of the same sunglass for Kanu is 1200 rupees, and now he sold it to Meenu at a 20% loss. Hence, the selling price of the sunglass now is $(1200 * 0.8) = 960$ rupees.

The cost price of the same sunglass for Meenu when he purchased it for the second time was 960 rupees. Now Meenu sold it Tanu, at a certain price such that the total profit of Meenu becomes 500 rupees.

Hence, on the second transaction (selling it to Tanu), Meenu made a profit of $(500-200) = 300$ rupees.

Hence, the profit made by Minu in the second transaction is $(300/960)*100\% = 31.25\%$

The correct option is C

 VIDEO SOLUTION

42. 407

Let the number of white shirts be m , and the number of blue shirts be n . Hence, the total cost of the shirts = $(1000m+1125n)$, and the number of shirts is $(m+n)$

The average price of the shirts is $\frac{1000m+1125n}{m+n}$.

It is given that he set a fixed market price which was 25% higher than the average cost of all the shirts. He sold all the shirts at a discount of 10%.

Hence, the average selling price of the shirts = $\left(\frac{1000m+1125n}{m+n}\right) \times \frac{5}{4} \times \frac{9}{10} = \frac{9}{8} \left(\frac{1000m+1125n}{m+n}\right)$

The average profit of the shirts = $\frac{9}{8} \left(\frac{1000m+1125n}{m+n}\right) - \frac{1000m+1125n}{m+n} = \frac{1}{8} \left(\frac{1000m+1125n}{m+n}\right)$

The total profit of the shirts = $\frac{1}{8} \left(\frac{1000m+1125n}{m+n}\right) \times (m+n) = \frac{1}{8} (1000m + 1125n)$

Now, $\Rightarrow \frac{1}{8} (1000m + 1125n) = 51000$

$\Rightarrow 1000m + 1125n = 51000 \times 8 = 408000$

Now to get the maximum number of shirts, we need to minimize n (since the coefficient of n is greater than the coefficient of m), but it can't be zero. Therefore, m has to be maximum.

$$m = \frac{408000-1125n}{1000}$$

The maximum value of m such that m , and both are integers is $m = 399$, and $n = 8$ (by inspection)

Hence, the maximum number of shirts = $m+n = 399+8 = 407$

 VIDEO SOLUTION

43. 2160

Let the cost price of the item be C

We are given that Bina sells this at 19% loss or at $(1 - 0.19)C = 0.81C$ at 4860

This gives us the value of C at Rs. 6000

If Bina had sold this at 17% profit, the selling price would have been $1.17 \times 6000 = 7020$

So Shyam bought the product at 4860 and sold it to Hari at 7020

Giving the profit made by Shyam to be $7020 - 4860 = 2160$

Therefore, 2160 is the correct answer.

 VIDEO SOLUTION

44. B

Let us fix the Cost Price of the product to be X , and the Selling Price of the product to be $1.4X$, since it is given that it is fixed to have a profit of 40%.

If the CP has been 40% less, making the CP $0.6X$,

And the selling price is 5 rupees less, making it $1.4X-5$

Profit will be 50%,

So, $1.5(0.6X) = 1.4X - 5$

$$0.9X = 1.4X - 5$$

$$0.5X = 5$$

$$X = 10$$

Original selling price will be 14.

 VIDEO SOLUTION

45. A

Let us say the cost price of an item is X

It is said that it is marked to make a profit of 20%.

That means it is marked at 1.2X

Ravi gets a 10% discount on the marked price,

$$0.9(1.2X) = 1.08X$$

Saves 15 rupees, so $1.2X - 1.08X$

$$0.12X = 15$$

$$X = 125$$

Profit made by Gopi is $0.08(125) = 10$ rupees.

 VIDEO SOLUTION



46. C

We are told that there was two successive increments in the salary, with the second increment percentage twice the first one. Total Increment was 187.5%.

Drawing up the equation

$$(1 + z)(1 + 2z) = 1.875$$

$$1 + 3z + 2z^2 = 1.875$$

$$2z^2 + 3z - 0.875$$

$$z = \frac{(-3 \pm \sqrt{9 + 8(0.875)})}{4}$$

$$z = \frac{(-3 \pm 4)}{4}$$

$$z = 0.25$$

Answer is 25%.

 VIDEO SOLUTION

47. 175

Cost price of each chair = 100

For 13 chairs, total cost = $13 \times 100 = 1300$

Profit = 26%, so total revenue

$$1.26 \times 1300 = 1638$$

We were told that this amount is equal to the discounted price of 12 chairs. So the discounted selling price per

$$\text{chair} = SP_{\text{disc}} = \frac{1638}{12} = 136.5$$

Discount offered = 22%, so:

$$MP = \frac{136.5}{0.78} = 175$$

VIDEO SOLUTION

48. B

The total number of products is $120 + 135 + 150 + 165 = 570$. Therefore, the total cost must have been $570 \times 240 = 136800$

Let the marked price for the products be X . The January, February, March, and April, selling prices would respectively be $0.8X$, $0.9X$, $0.95X$, and X ; based on the discounts given on each of the months.

The revenue earned from selling the products in each of the months would be: $120 * 0.8X + 135 * 0.9X + 150 * 0.95X + 165 * X = 525X$.

The profit earned would be: Revenue - Cost, and therefore,

$$525X - 136800 = 138825$$

$$525X = 275625$$

$$X = \frac{275625}{525} = 525$$

The correct answer is option B, Rs. 525.

VIEW SOLUTION

49. C

Sure! Here's a clean solution:

Let the marked price be M and the initial discount rate be d . The cost price is 1650. A profit of 20% means the selling price is

$$SP = 1650 \times 1.2 = 1980$$

With discount d ,

$$M(1 - d) = 1980$$

If the discount is doubled, the selling price becomes $(M(1-2d))$, and the profit is 110, so

$$M(1 - 2d) - 1650 = 110 \Rightarrow M(1 - 2d) = 1760$$

Subtracting the two equations:

$$M(1 - d) - M(1 - 2d) = 1980 - 1760 \Rightarrow Md = 220 \Rightarrow M = \frac{220}{d}$$

Plug into the first equation:

$$\frac{220}{d}(1 - d) = 1980 \Rightarrow \frac{1-d}{d} = 9 \Rightarrow 1 - d = 9d \Rightarrow d = 0.1$$

So the initial discount rate is 10% and the marked price is

$$M = \frac{220}{0.1} = 2200$$

Now, let the discount rate be (r) such that the profit percentage equals the discount percentage. Then

$$\frac{2200(1-r)-1650}{1650} = r$$

Simplify:

$$2200(1 - r) - 1650 = 1650r \Rightarrow 2200 - 2200r - 1650 = 1650r \Rightarrow 550 = 3850r \Rightarrow r = \frac{550}{3850} \approx 0.142857$$

Thus, the required discount rate is approximately 14%

VIDEO SOLUTION

50. C

It is given that the maximum cost is 12 lakhs

$1200000 \leq T \times 10000 + V \times 15000$ (Where T is number of TV's and V is number of VCR's)

and as we know $T + V = 100$

By solving the above two equation, we will get $v \leq 40$

We have to keep the no. of VCRs as maximum, as they fetch higher profit than TV, so

$$T = 60$$

$$V = 40$$

 VIDEO SOLUTION



51. B

As we know for maximizing profit, Total Revenue = Marginal Cost

$$\text{i.e. } T \times 10000 + V \times 15000 = 1200000$$

$$\text{and } T + V = 120$$

By solving above two equations we will get

$$T = 120$$

$$V = 0$$

Hence ratio will be 0.

 VIDEO SOLUTION

52. A

As from his original cost values, number of TV's were 60 and VCR's were 40

$$\text{And now total selling price will be} = 12200 \times 60 + 18300 \times 40 = 1464000$$

Hence profit = selling price - cost price

$$= 1464000 - 1200000$$

$$= 264000 \text{ rs.}$$

 VIDEO SOLUTION

53. B

The total property consists of cash, gold coins and silver bars.

And ghosh babu gave equal parts to 4 daughters, hence they should have 25% of total property each.

As eldest daughter possess gold coins as 20% worth of total property, so 25000 cash should be equal to 5% of total property.

$$\text{So total property will be} = \frac{25000 \times 100}{5} = 500000$$

Hence property amounting only gold coins and silver bars will be = Total property - Total Cash

$$\text{i.e.} = 500000 - (25000 + 50000 + 75000 + 75000)$$

$$= 2,75,000$$

 VIDEO SOLUTION

54. A

The total property consists of cash, gold coins and silver bars.

And ghosh babu gave equal parts to 4 daughters, hence they should have 25% of total property each.

As eldest daughter possess gold coins as 20% worth of total property, so 25000 cash should be equal to 5% of total property.

$$\text{So total property will be} = \frac{25000 \times 100}{5} = 500000$$

 VIDEO SOLUTION

55. D

Let the number of gold bars be $7a$ and the number of silver bars be $27a$.

The total value of the gold and silver bars is $500000 - 225000 = 275000$

Therefore, $4000 \times 7a + 1000 \times 27a = 275000$

Or, $55000a = 275000$

Or, $a = 5$

Therefore the number of silver bars with Ghosh Babu is $27a = 135$

 VIDEO SOLUTION

A banner for 'CAT Daily Targets' featuring two people on the left and a calendar icon on the right. The text 'CAT Daily Targets' is in large bold letters, and below it, a yellow box says 'Step by step, heading towards 99+ percentile'. The 'cracku' logo is in the top right corner.

CAT Daily Targets
Step by step, heading towards 99+ percentile

56. B

Cost price per watch = 150

Cost price for 1500 watches = $1500 \times 150 = 225000$

Total expense = $225000 + 30000 = 255000$

Selling price for season = $1200 \times 250 = 300000$

For out of season = $300 \times 100 = 30000$

Total selling = $300000 + 30000 = 330000$

Profit = $330000 - 255000 = 75000$

 VIDEO SOLUTION

57. B

Break even implies that cost price is equal to selling price

Hence let's say in season x watches were sold

Cost price will be = $1500 \times 150 + 30000 = 255000$

So total selling price = $250x + (1500 - x)100 = 255000$

Or $x = 700$

 VIDEO SOLUTION

58. A

Let the total production cost be 100.

Hence, selling price is 120.

Price of German component A is 30 and the price of the US component B is 50

After change in exchange rate, price of German component is $30 \times 1.3 = 39$

and price of US component is $50 \times 1.22 = 61$

Total increase equals $39 + 61 - 30 - 50 = 20$

Hence, the minimum production cost is $100 + 20 = 120$

The maximum possible selling price is $120 \times 110\% = 132$.

So, maximum possible gain is $(132 - 120) / 120 = 10\%$

 VIEW SOLUTION

59. **B**

Let the total production cost be 100.

Hence, selling price is 120.

Price of German component A is 30 and the price of the US component B is 50

After change in exchange rate, price of German component is $30 \times 1.2 = 36$

and price of US component is $50 \times 0.88 = 44$

Total increase equals $(36 + 44) - (30 + 50) = 0$

Hence, the total production cost did not change.

As the selling price also did not change, the gain percentage equals 20%

[VIEW SOLUTION](#)

60. **A**

After the first game, Ghosh Babu picked 8 of clubs => He gets Rs 8. Then the dealer picked Queen of clubs => Ghosh Babu pays Rs 16 => Ghosh Babu is at a loss of Rs 8 after 1st game.

After the second game, Ghosh Babu picked 10 of hearts => He gets Rs 10. Then the dealer picked 2 of spades => Ghosh Babu gets another Rs 10 => Ghosh Babu is now at a profit of Rs 12.

After the third game, Ghosh Babu picked six of diamonds => He gets Rs 6. Then the dealer picked ace of hearts => Ghosh Babu pays Rs 6 to dealer => Ghosh Babu is still at a profit of Rs 12.

In the fourth game, Ghosh Babu picks 8 of spades => He gets Rs 8. The the dealer picks jack of spades => Ghosh Babu pays Rs 16 to dealer => Ghosh Babu is at a profit of Rs 4.

Hence, the maximum profit earned is Rs 12.

[VIEW SOLUTION](#)



61. **B**

After the first game, Ghosh Babu picked 8 of clubs => He gets Rs 8. Then the dealer picked Queen of clubs => Ghosh Babu pays Rs 16 => Ghosh Babu is at a loss of Rs 8 after 1st game.

After the second game, Ghosh Babu picked 10 of hearts => He gets Rs 10. Then the dealer picked 2 of spades => Ghosh Babu gets another Rs 10 => Ghosh Babu is now at a profit of Rs 12.

After the third game, Ghosh Babu picked six of diamonds => He gets Rs 6. Then the dealer picked ace of hearts => Ghosh Babu pays Rs 6 to dealer => Ghosh Babu is still at a profit of Rs 12.

In the fourth game, Ghosh Babu picks 8 of spades => He gets Rs 8. The the dealer picks jack of spades => Ghosh Babu pays Rs 16 to dealer => Ghosh Babu is at a profit of Rs 4.

Hence, the maximum profit earned is Rs 12.

The maximum loss that Ghosh Babu had was Rs 8.

He must have had at least Rs 8 so that he did not have to borrow any amount from others.

[VIEW SOLUTION](#)

62. **D**

After the first game, Ghosh Babu picked 8 of clubs => He gets Rs 8. Then the dealer picked Queen of clubs => Ghosh Babu pays Rs 16 => Ghosh Babu is at a loss of Rs 8 after 1st game.

After the second game, Ghosh Babu picked 10 of hearts => He gets Rs 10. Then the dealer picked 2 of spades => Ghosh Babu gets another Rs 10 => Ghosh Babu is now at a profit of Rs 12.

After the third game, Ghosh Babu picked six of diamonds => He gets Rs 6. Then the dealer picked ace of hearts => Ghosh Babu pays Rs 6 to dealer => Ghosh Babu is still at a profit of Rs 12.

In the fourth game, Ghosh Babu picks 8 of spades => He gets Rs 8. The the dealer picks jack of spades => Ghosh Babu pays Rs 16 to dealer => Ghosh Babu is at a profit of Rs 4.

As Ghosh Babu earned a profit of Rs 4 and now he has Rs 100, he initially would have had $100 - 4 = \text{Rs } 96$.

 [VIEW SOLUTION](#)